

- Q.28 Explain the advantages of disadvantages of an axial flow compressor
- Q.29 What are the three types of combustion chambers?
- Q.30 Explain the engine ground testing procedure
- Q.31 Write down the factors which affect the performance of combustion chamber.
- Q.32 What are the disadvantages of a centrifugal compressor?
- Q.33 What do you mean by after burning in a jet engine?
- Q.34 What is the main difference between piston engine and turbine engine.
- Q.35 What are the different methods of thrust augmentation?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain ideal Brayton cycle, real cycle, cycle with intercooler and cycle with reheat.
- Q.37 Explain the various parts of a gas turbine engine. Also describe the salient features of each type.
- Q.38 Discuss the complete air system of a turbine with a diagram.

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5th Sem / AME

Subject:- Turbo Propeller and Turbo Jet Engine - I

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The thermodynamic cycle for gas turbine engine is _____
- Brayton cycle
 - Stirling Cycle
 - Rankine cycle
 - Reverse Brayton cycle
- Q.2 Gas turbines are used in aircraft propulsion because
- they are light
 - They are compact
 - they have high power-to-weight ratio
 - all of the mentioned
- Q.3 The propulsive efficiency is maximum at
- $V=V_j$
 - $V=.5 V_j$
 - $V<V_j$
 - $V>V_j$
- Q.4 _____ compressors can be used in turbojets
- Axial
 - Centrifugal
 - Axial & centrifugal
 - None of the above

- Q.5 By pass ratio, modern turbofan engine is of the order of.
- a) 1 to 2 b) 4 to 10
c) 10 to 20 d) 20 to 40
- Q.6 What is the ratio of air and fuel injected into the combustion chamber?
- a) ~100 b) ~50
c) ~15 d) ~5
- Q.7 Extra fuel is injected to after burners to _____ the thrust.
- a) decrease
b) increase
c) extra fuel is not injected
d) none of the mentioned
- Q.8 The bleed air is used for
- a) Air conditioning b) Pneumatic system
c) Both a and b d) none of the above
- Q.9 The latest development in turbine blades is the introduction of
- a) Titanium b) Ceramic
c) Chromium d) None of the above
- Q.10 Which of the following is an example of a cooling system?
- a) Downdraft cooling b) Downdraft lofting
c) Sideway lofting d) Sideway circulation

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The process in the combustion chamber is _____ process.
- Q.12 What do you mean by a jet?
- Q.13 What is the purpose of tertiary air?
- Q.14 Inlet of a gas turbine engine act as nozzle or diffuser?
- Q.15 What are different types of compressors?
- Q.16 What is the effect of by-pass air on performance?
- Q.17 How turbine blades are cooled?
- Q.18 Define equivalence ratio.
- Q.19 What is a free turbine?
- Q.20 What type of flow is in the vane less space of centrifugal compressor?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a brief note on a turbine engine.
- Q.22 Draw a typical reheat turbojet cycle.
- Q.23 Write a brief note on different types of nozzles used.
- Q.24 What is the purpose of multispooling?
- Q.25 How are combustion chambers constructed?
- Q.26 What is the use of gears in gas turbine engine?
- Q.27 What are the important factors affecting combustor design?